

# H524 Introduction to Biostatistics

## Week 8: Contingency Tables and Chi-Square Tests

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# Outline

- 1 Review: Where We've Been
- 2 The Origins: Fisher's Tea Experiment
- 3 Fisher's Exact Test
- 4 Chi-Square Goodness-of-Fit Test (Ch 15.1)
- 5 Contingency Tables (Ch 15.2-15.3)
- 6 Practical Considerations and R Implementation
- 7 Preview: Week 9 - Effect Sizes (Optional)

# Quick Review: Weeks 6-7 Recap

## Recent Topics Covered:

- **Week 6:** Hypothesis Testing Framework
  - One-sample and two-sample t-tests
  - Paired tests
  - Type I and Type II errors, power
- **Week 7:** Nonparametric Methods
  - Sign test, Wilcoxon signed-rank test
  - Mann-Whitney U test
  - When to use nonparametric vs parametric tests

**Common theme:** Comparing means (continuous outcomes)

**This week:** Categorical outcomes and associations

# From Continuous to Categorical Data

## So far in H524:

- Continuous outcomes: blood pressure, cholesterol, pain scores
- Methods: t-tests, confidence intervals, ANOVA
- Questions: Are means different between groups?

## New challenge: Categorical outcomes

- Binary: disease/no disease, treatment success/failure
- Nominal: blood type (A, B, AB, O), race/ethnicity
- Ordinal: disease stage (I, II, III, IV), pain level (none, mild, moderate, severe)
- Count data: number of deaths, adverse events, cases

## Key Questions:

- Is there an association between two categorical variables?
- Do observed frequencies match expected frequencies?
- Are two categorical variables independent?

# The Lady Tasting Tea (1920s)

## The Claim:

- Dr. Muriel Bristol, a phycologist at Rothamsted Experimental Station (England)
- Claimed she could taste whether milk was poured before or after tea
- Most people thought this was impossible

## The Skeptic:

- Sir Ronald A. Fisher, famous English statistician, doubted her claim
- Proposed testing her claim through a controlled experiment
- Designed an experiment that would become legendary

## A Foundational Moment in Statistics:

- Fisher used this as the famous example for his exact test
- Original exposition of the null hypothesis concept (1935)
- Established principles for randomized experiments and hypothesis testing

*Note: Rothamsted Experimental Station, England, early 1920s*

# Fisher's Experimental Design

## The Setup:

- Prepare 8 cups of tea
- 4 cups: milk poured first, then tea ( $M \rightarrow T$ )
- 4 cups: tea poured first, then milk ( $T \rightarrow M$ )
- Present all 8 cups in random order

## The Task:

- Dr. Bristol must identify which 4 cups are milk-first
- She knows there are exactly 4 of each type
- No other cues allowed (temperature, color controlled)

## The Question:

- If she's just guessing randomly, what's the probability she gets all 4 correct?
- This is a question about **counting combinations**

# Calculating the Probability

## Total possible outcomes:

- How many ways can you choose 4 cups from 8?
- $\binom{8}{4} = \frac{8!}{4! \times 4!} = 70$  different combinations

## Favorable outcomes (all correct):

- She correctly identifies all 4 milk-first cups
- Only 1 way to do this:  $\binom{4}{4} \times \binom{4}{0} = 1$

## Probability by pure chance:

$$p = \frac{1}{70} = 0.0143$$

## The Result:

- Dr. Bristol correctly identified all 8 cups!
- Probability of this happening by chance:  $p = 0.0143$  (less than 5%)
- Conclusion: Strong evidence she really could taste the difference

# Who Was Sir Ronald A. Fisher?

**Sir Ronald Aylmer Fisher (1890-1962)**

**Prominent English Statistician and Geneticist**

**Foundational Contributions to Statistics:**

- **Experimental design:** Randomization, blocking, factorial designs
- **Analysis of Variance (ANOVA):** Comparing multiple groups
- **Maximum likelihood estimation:** Parameter estimation method
- **Fisher's exact test:** Born from the tea experiment
- **Genetics:** Reconciled Mendelian genetics with Darwinian evolution

**Career:**

- Worked at Rothamsted Experimental Station (1919-1933)
- Later: Professor at University College London and Cambridge
- One of the most influential scientists of the 20th century

*Reference: Fisher, R.A. (1935). The Design of Experiments. Oliver & Boyd.*

# Fisher's Exact Test: Hypotheses

For a  $2 \times 2$  contingency table:

|       | Col 1 | Col 2 |
|-------|-------|-------|
| Row 1 | $a$   | $b$   |
| Row 2 | $c$   | $d$   |

**Hypotheses:**

- $H_0$ : The two variables are independent (no association)
- $H_A$ : The two variables are associated (dependent)

**What the p-value tells us:**

- Probability of seeing a table this extreme (or more) if  $H_0$  is true
- Small p-value  $\Rightarrow$  reject  $H_0 \Rightarrow$  evidence of association
- Large p-value  $\Rightarrow$  fail to reject  $H_0 \Rightarrow$  no evidence of association

# Fisher's Exact Test: The Math

**For a  $2 \times 2$  table:**

|       | Col 1   | Col 2   | Total   |
|-------|---------|---------|---------|
| Row 1 | $a$     | $b$     | $a + b$ |
| Row 2 | $c$     | $d$     | $c + d$ |
| Total | $a + c$ | $b + d$ | $n$     |

**Exact probability (hypergeometric distribution):**

$$P = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}}$$

**The idea:**

- Calculate probability of observed table (given fixed margins)
- Calculate probabilities of all more extreme tables
- Sum these probabilities = exact p-value
- No approximation needed!

**Note:** Hand calculation tedious; R does it for us

# Fisher's Exact Test: Example with Tea Data

Representing Dr. Bristol's results as a  $2 \times 2$  table:

|                              | Actually $M \rightarrow T$ | Actually $T \rightarrow M$ | Total |
|------------------------------|----------------------------|----------------------------|-------|
| Identified $M \rightarrow T$ | 4                          | 0                          | 4     |
| Identified $T \rightarrow M$ | 0                          | 4                          | 4     |
| Total                        | 4                          | 4                          | 8     |

R code:

```
tea <- matrix(c(4, 0, 0, 4), nrow=2)
fisher.test(tea)
```

**Result:**  $p = 0.0143$

**Conclusion:** Strong evidence she could really taste the difference!

# Fisher's Exact Test: Modern Medical Example

**Scenario:** Rare disease study with small sample

|             | Disease | No Disease | Total |
|-------------|---------|------------|-------|
| Exposed     | 8       | 2          | 10    |
| Not Exposed | 3       | 7          | 10    |
| Total       | 11      | 9          | 20    |

**R code (replace numbers with your data):**

```
data <- matrix(c(8, 2, 3, 7), nrow=2)
fisher.test(data)
```

**Result:** P-value = 0.035 (two-sided)

**Conclusion:** Significant association between exposure and disease

**When to use:** Small samples (especially when any cell count  $< 5$ )

# Scaling Up: The Challenge of Large Samples

**Fisher's exact test works beautifully for small samples...**

**But what about large datasets?**

- Tea experiment: 8 cups  $\rightarrow$  70 combinations (countable)
- Medical study: 100 patients  $\rightarrow$  thousands of combinations
- Large survey: 1000 participants  $\rightarrow$  billions of combinations!

**The Problem:**

- Counting all combinations becomes computationally expensive
- Even modern computers struggle with very large samples
- We need a more efficient approach

**The Solution:**

- For large samples, we can use a mathematical approximation
- The hypergeometric distribution approximates the chi-square distribution
- This gives us the **chi-square test**

# Introduction to Chi-Square Tests

## The chi-square ( $\chi^2$ ) test family:

- Tests for categorical data
- Based on comparing observed vs expected frequencies
- Large-sample approximation to Fisher's exact test

## Two main types:

- 1 **Goodness-of-fit test:** Does a single categorical variable follow a specified distribution?
  - Example: Are blood types distributed as expected (45% O, 40% A, 11% B, 4% AB)?
- 2 **Test of independence:** Are two categorical variables associated?
  - Example: Is smoking status associated with lung cancer?

**Reading:** Pagano & Gauvreau Chapters 15-16

# Goodness-of-Fit Test: The Basic Idea

**Question:** Do observed data match a theoretical distribution?

**Setup:**

- One categorical variable with  $k$  categories
- Null hypothesis specifies expected proportions:  $p_1, p_2, \dots, p_k$
- Observed counts:  $O_1, O_2, \dots, O_k$  (total  $n$ )
- Expected counts:  $E_i = np_i$  for each category

**Chi-square test statistic:**

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

**Intuition:**

- If observed  $\approx$  expected, then  $\chi^2$  is small
- If observed differs from expected, then  $\chi^2$  is large
- Large  $\chi^2 \Rightarrow$  reject  $H_0$  (poor fit)

# Chi-Square Distribution

## Properties of the $\chi^2$ distribution:

- Right-skewed, non-negative values only
- Determined by degrees of freedom:  $df = k - 1$
- Mean =  $df$ , Variance =  $2 \times df$
- As  $df$  increases, approaches normal distribution

## Assumptions for chi-square tests:

- 1 Random sample
- 2 Independent observations
- 3 Expected counts  $\geq 5$  in each category
  - If violated: combine categories or use Fisher's exact test
- 4 Sufficiently large sample size

**R function:** `pchisq(q, df, lower.tail=FALSE)` for p-values

# Example 1: Blood Type Distribution

**Scenario:** U.S. population blood type distribution:

- Type O: 45%
- Type A: 40%
- Type B: 11%
- Type AB: 4%

**Question:** Does a sample of 200 hospital patients match this distribution?

**Observed counts:**

| Blood Type | O  | A  | B  | AB |
|------------|----|----|----|----|
| Observed   | 85 | 78 | 28 | 9  |

**Hypotheses:**

- $H_0$ : Hospital patients have same blood type distribution as general population
- $H_A$ : Distribution differs from general population

## Example 1: Calculations

### Step 1: Calculate expected counts

Expected:  $E_i = n \times p_i$  where  $n = 200$

| Blood Type         | O  | A  | B  | AB |
|--------------------|----|----|----|----|
| Observed ( $O_i$ ) | 85 | 78 | 28 | 9  |
| Expected ( $E_i$ ) | 90 | 80 | 22 | 8  |

### Step 2: Calculate chi-square statistic

$$\begin{aligned}\chi^2 &= \frac{(85 - 90)^2}{90} + \frac{(78 - 80)^2}{80} + \frac{(28 - 22)^2}{22} + \frac{(9 - 8)^2}{8} \\ &= \frac{25}{90} + \frac{4}{80} + \frac{36}{22} + \frac{1}{8} \\ &= 0.278 + 0.050 + 1.636 + 0.125 = 2.089\end{aligned}$$

## Example 1: Conclusion

### Step 3: Find p-value

- Degrees of freedom:  $df = k - 1 = 4 - 1 = 3$
- Test statistic:  $\chi^2 = 2.089$
- P-value:  $P(\chi^2_3 > 2.089) = 0.554$

### R code:

```
pchisq(2.089, df=3, lower.tail=FALSE)  
# Returns: 0.554
```

### Conclusion:

- P-value = 0.554 > 0.05
- Fail to reject  $H_0$
- No evidence that hospital blood type distribution differs from general population

# Contingency Tables: Introduction

**Contingency table** (cross-tabulation): Displays frequency distribution of two categorical variables

**General  $r \times c$  table:**

- $r$  rows (levels of variable 1)
- $c$  columns (levels of variable 2)
- Cell counts:  $O_{ij}$  = observed count in row  $i$ , column  $j$
- Row totals, column totals, grand total  $n$

**Key Question:** Are the two variables independent?

**Hypotheses:**

- $H_0$ : Row and column variables are independent (no association)
- $H_A$ : Row and column variables are associated (dependent)

# $2 \times 2$ Contingency Table

**Most common case:** Two binary variables

|             | Disease | No Disease | Total   |
|-------------|---------|------------|---------|
| Exposed     | $a$     | $b$        | $a + b$ |
| Not Exposed | $c$     | $d$        | $c + d$ |
| Total       | $a + c$ | $b + d$    | $n$     |

## Examples:

- Exposure (smoking) vs Disease (lung cancer)
- Treatment (drug vs placebo) vs Outcome (success vs failure)
- Risk factor (hypertension) vs Event (stroke vs no stroke)

## Measures we can calculate:

- Relative risk
- Odds ratio
- Chi-square test for independence

# Where Do Expected Values Come From?

**Goal:** Find expected count in cell  $a$  (Exposed & Disease) if independent.

**Under independence:**

$$P(\text{Exposed AND Disease}) = P(\text{Exposed}) \times P(\text{Disease})$$

**From the  $2 \times 2$  table:**

- $P(\text{Exposed}) = \frac{a+b}{n}$  (row total)
- $P(\text{Disease}) = \frac{a+c}{n}$  (column total)

**Expected count in cell  $a$ :**

$$E_a = n \times \frac{a+b}{n} \times \frac{a+c}{n} = \frac{(a+b)(a+c)}{n}$$

**General formula:**

$$E_{ij} = \frac{(\text{Row total}) \times (\text{Column total})}{n}$$

**Key insight:** Expected values show what we'd see if variables were independent.

# Chi-Square Test of Independence: Hypotheses

## Testing association between two categorical variables

### Hypotheses:

- $H_0$ : The row and column variables are independent (no association)
- $H_A$ : The row and column variables are associated (dependent)

### What the p-value tells us:

- If  $p\text{-value} < \alpha$  (typically 0.05): Reject  $H_0$ 
  - Evidence that variables are associated
  - Observed counts differ significantly from what we'd expect under independence
- If  $p\text{-value} \geq \alpha$ : Fail to reject  $H_0$ 
  - No evidence of association
  - Observed counts consistent with independence

**Note:** Chi-square test tells us IF there's an association, not HOW STRONG it is (use effect sizes for that)

# Chi-Square Test of Independence: Formula

**Test statistic for  $r \times c$  table:**

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

**Expected count formula:**

$$E_{ij} = \frac{(\text{Row } i \text{ total}) \times (\text{Column } j \text{ total})}{\text{Grand total}}$$

**Degrees of freedom:**

$$df = (r - 1) \times (c - 1)$$

**Interpretation:**

- Expected counts assume independence (null hypothesis)
- Large differences between  $O_{ij}$  and  $E_{ij} \Rightarrow$  evidence of association
- Compare  $\chi^2$  to chi-square distribution with appropriate  $df$

## Example 3: Smoking and Lung Cancer

**Case-control study:** 100 lung cancer patients, 100 controls

|            | Lung Cancer | No Cancer | Total |
|------------|-------------|-----------|-------|
| Smoker     | 85          | 55        | 140   |
| Non-smoker | 15          | 45        | 60    |
| Total      | 100         | 100       | 200   |

**Question:** Is there an association between smoking and lung cancer?

**Hypotheses:**

- $H_0$ : Smoking status and lung cancer are independent
- $H_A$ : Smoking status and lung cancer are associated

## Example 3: Expected Counts

Calculate expected counts under independence:

$$E_{ij} = \frac{(\text{Row total}) \times (\text{Column total})}{n}$$

|            | Cancer                            | No Cancer                         |
|------------|-----------------------------------|-----------------------------------|
| Smoker     | $\frac{140 \times 100}{200} = 70$ | $\frac{140 \times 100}{200} = 70$ |
| Non-smoker | $\frac{60 \times 100}{200} = 30$  | $\frac{60 \times 100}{200} = 30$  |

Summary table:

| Cell                   | Observed | Expected |
|------------------------|----------|----------|
| Smoker + Cancer        | 85       | 70       |
| Smoker + No Cancer     | 55       | 70       |
| Non-smoker + Cancer    | 15       | 30       |
| Non-smoker + No Cancer | 45       | 30       |

## Example 3: Chi-Square Calculation

Observed counts with expected in parentheses:

|            | Cancer  | No Cancer |
|------------|---------|-----------|
| Smoker     | 85 (70) | 55 (70)   |
| Non-smoker | 15 (30) | 45 (30)   |

Calculate  $\chi^2$ :

$$\begin{aligned}\chi^2 &= \sum \frac{(O - E)^2}{E} = \frac{(85 - 70)^2}{70} + \frac{(55 - 70)^2}{70} + \frac{(15 - 30)^2}{30} + \frac{(45 - 30)^2}{30} \\ &= 3.21 + 3.21 + 7.50 + 7.50 = 21.43\end{aligned}$$

$df = 1$ ,  $p\text{-value} < 0.0001 \Rightarrow$  **Very strong evidence of association**

## Example 4: Treatment Efficacy

**Clinical trial:** New drug vs placebo for treating infection

**Observed counts with expected in parentheses:**

|         | Cured   | Not Cured | Total |
|---------|---------|-----------|-------|
| Drug    | 48 (40) | 12 (20)   | 60    |
| Placebo | 32 (40) | 28 (20)   | 60    |
| Total   | 80      | 40        | 120   |

**Calculate  $\chi^2$ :**

$$\chi^2 = \frac{(48 - 40)^2}{40} + \frac{(12 - 20)^2}{20} + \frac{(32 - 40)^2}{40} + \frac{(28 - 20)^2}{20} = 9.60$$

$df = 1$ ,  $p\text{-value} = 0.002 \Rightarrow$  **Strong evidence drug is more effective**

# Larger Contingency Tables

**Example:**  $3 \times 3$  table - Disease severity by age group

|                | Mild | Moderate | Severe | Total |
|----------------|------|----------|--------|-------|
| Young (< 40)   | 45   | 20       | 5      | 70    |
| Middle (40-60) | 30   | 35       | 15     | 80    |
| Older (> 60)   | 15   | 25       | 10     | 50    |
| Total          | 90   | 80       | 30     | 200   |

## Analysis:

- $df = (3 - 1)(3 - 1) = 4$
- Calculate 9 expected counts
- Sum contributions:  $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
- Compare to  $\chi^2_4$  distribution

**Interpretation:** Significant result indicates association between age and disease severity, but doesn't specify pattern

# Advanced Topic: McNemar's Test for Paired Data

## A specialized test for paired/matched binary data

### When is it used?

- Same subjects measured twice (before/after studies)
- Matched pairs (twins, matched case-control designs)
- Any situation where observations are paired

### Why different from chi-square?

- Chi-square assumes independent observations
- McNemar's accounts for the pairing/matching
- Uses only discordant pairs (subjects who changed)

### For this course:

- More advanced test - covered in detail during lab
- Worksheet Problem 5 is optional practice
- Important when you encounter paired data in research

# Chi-Square Tests in R (Part 1 - Goodness-of-Fit)

## Goodness-of-fit test:

```
# Observed counts
observed <- c(85, 78, 28, 9)

# Expected proportions
expected_props <- c(0.45, 0.40, 0.11, 0.04)

# Chi-square test
chisq.test(observed, p=expected_props)
```

## Output includes:

- $\chi^2$  statistic
- Degrees of freedom
- P-value

# Chi-Square Tests in R (Part 2 - Test of Independence)

## Test of independence:

```
# Create contingency table
data <- matrix(c(85, 55, 15, 45), nrow=2, byrow=TRUE)
rownames(data) <- c("Smoker", "Non-smoker")
colnames(data) <- c("Cancer", "No Cancer")

# Chi-square test
chisq.test(data)

# With continuity correction (for 2x2 tables)
chisq.test(data, correct=TRUE)
```

## Fisher's exact test:

```
# For 2x2 table  
data <- matrix(c(8, 2, 3, 7), nrow=2)  
fisher.test(data)
```

## When to use:

- Small sample sizes
- Expected counts  $< 5$
- Exact p-values needed

## Optional Preview of Next Week

### Coming in Week 9:

- Relative Risk (RR)
- Odds Ratio (OR)
- Interpreting effect sizes
- When to use RR vs OR

**Note:** The following slides are optional. Feel free to read ahead!

# Effect Sizes: Relative Risk and Odds Ratio (Week 9 Preview)

**These measures quantify the strength of association in 2×2 tables**

## **Relative Risk (RR):**

- Ratio of disease risk in exposed vs unexposed groups
- $RR = \frac{a/(a+b)}{c/(c+d)}$  (from 2×2 table)
- Interpretation:  $RR = 2.0$  means exposed group has twice the risk
- $RR = 1.0$  means no association

## **Odds Ratio (OR):**

- Ratio of odds of disease in exposed vs unexposed groups
- Odds in exposed =  $a/b$ , odds in unexposed =  $c/d$
- $OR = \frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$  (cross-product)
- Interpretation:  $OR = 2.0$  means exposed group has twice the odds
- $OR = 1.0$  means no association

**When to use:** RR for cohort studies, OR for case-control studies

## Odds ratio and confidence interval:

```
# Using fisher.test
result <- fisher.test(data)
result$estimate      # Odds ratio
result$conf.int      # 95% CI

# Manual calculation
OR <- (data[1,1] * data[2,2]) / (data[1,2] * data[2,1])
```

## Relative risk (manual):

```
RR <- (data[1,1]/sum(data[1,])) /
      (data[2,1]/sum(data[2,]))
```

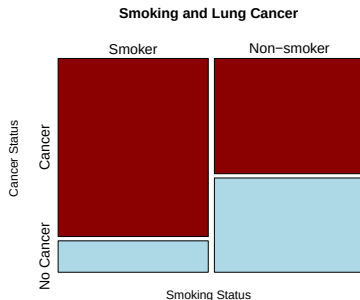
# Visualizing Contingency Tables (Part 1 - Mosaic Plot)

## R code:

```
# Create data
data <- matrix(c(85, 55, 15, 45),
               nrow=2)
rownames(data) <- c("Smoker",
                    "Non-smoker")
colnames(data) <- c("Cancer",
                    "No Cancer")

# Mosaic plot
mosaicplot(data,
            main="Smoking and Cancer",
            color=c("darkred",
                    "lightblue"),
            xlab="Smoking Status",
            ylab="Cancer Status")
```

## Output:



Area proportional to cell counts.  
Shows association visually.

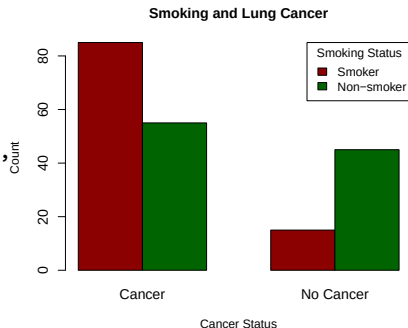
# Visualizing Contingency Tables (Part 2 - Counts)

## R code:

```
barplot(data, beside=TRUE,  
        legend=TRUE,  
        col=c("darkred",  
              "darkgreen"),  
        main="Smoking and Cancer",  
        xlab="Cancer Status",  
        ylab="Count")
```

Shows absolute counts side-by-side for comparison.

## Output:



# Visualizing Contingency Tables (Part 3 - Proportions)

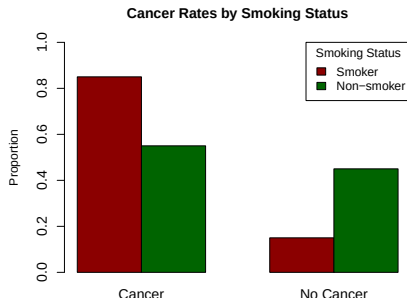
## R code:

```
# Row proportions
prop_data <- prop.table(data,
                          margin=1)

barplot(prop_data,
        beside=TRUE,
        legend=TRUE,
        col=c("darkred",
              "darkgreen"),
        main="Cancer Rates",
        ylab="Proportion")
```

Shows cancer rates within each smoking group.

## Output:



Easier to see association: smokers have higher cancer proportion.

# Common Mistakes and Pitfalls

## 1. Using chi-square with small expected counts

- Check all expected counts  $\geq 5$
- If violated: use Fisher's exact test or combine categories

## 2. Confusing association with causation

- Chi-square shows association, not causation
- May be confounding variables

## 3. Using chi-square for paired/matched data

- Independence assumption violated
- Use McNemar's test instead

## 4. Ignoring effect size

- Significant p-value doesn't mean strong association
- Always examine the actual counts and proportions
- Effect sizes (OR, RR) covered next week

## 5. Multiple testing without adjustment

- Testing many pairs in large tables inflates Type I error
- Consider Bonferroni or other corrections

# Choosing the Right Test

## **One categorical variable vs expected distribution**

- Chi-square goodness-of-fit

## **Two independent categorical variables (large sample)**

- Chi-square test of independence

## **Two independent categorical variables (small sample)**

- Fisher's exact test

## **Paired/matched binary data**

- McNemar's test

## **Ordinal variables with natural ordering**

- Consider Cochran-Armitage trend test

## **Multiple categorical variables**

- Log-linear models (beyond this course)